Course Syllabus

PHYS2325-Spring2026

Mechanics

Revised January 2026

Lamya Saleh

Course Number/Section	**PHYS 2325.003**
Course Title	**Mechanics**
Term	Spring 2026
Time & Location	Tues & Thurs 2:30 - 3:45 am, SCI 1.220
Instruction Mode: Traditional	

Instructor Contact Information
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Course Pre-requisites and Co-requisites
Prerequisite: MATH 2413 or MATH 2417
Corequisites: MATH 2414 or MATH 2419 and PHYS 2125

This course is an introductory course into Mechanics, covering basic physics concepts. As a content oriented course, be prepared to invest time into completing the assigned work on a weekly basis. It is understandable that finding the right balance as a college student can be challenging. This thought has been on our minds when designing this course. We employ a range of teaching methods to make it easier for learners with different backgrounds and different learning styles. We are here to help you succeed. Please do not hesitate to ask for help if you need it.
Welcome to the class!

Course Description:
Mechanics is the first of the two required freshmen calculus-based physics courses. The course will be taught at the level introduced in introductory physics textbooks. Calculus based.
Basic physics including a study of space and time, kinematics, forces, energy and momentum, conservation laws, rotational motion, torques, and harmonic oscillation. Two lectures per week.

Class participation: Regular class participation is expected. Students who fail to participate in class regularly are inviting scholastic difficulty. Successful participation is defined as consistently adhering to University requirements, as presented in this syllabus. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Do I need to purchase a Textbook?

No need to purchase a textbook. The curriculum for this course is tailored to the needs of our students. Basic concepts required will be available to you on Elearning while retrieval and practice exercises will be covered in class. You are encouraged to use a textbook of your choice as a reference. Any relatively recent Introductory physics textbook will be sufficient. Here are a few suggestions.

University Physics, Young and Freedman, Volume 1;
Fundamentals of Physics, D. Halliday, R. Resnick, J. Walker. Volume 2;
Physics for Scientists and Engineers, R. Serway & J. Jewett;
Physics for Engineers & Scientists , H. Ohanion & J. Markert.

We will be using "University Physics" by Young and Freedman for our Mastering assignments.

Do I need to purchase access to Mastering Physics ?

Access to Mastering Physics *is a requirement* for this course. The mastering version of this course is titled: **Mechanics-Spring26**

Course ID: saleh91294 (Note: ***you should not need this ID#*** to register or to access the assignments)

Instructions on how to purchase access to Mastering Physics:

- First, access the PHYS2325 class on E-learning. This course can be accessed using your UT Dallas NetID account on https://elearning.utdallas.edu/ website.

- UT Dallas provides eLearning technical support 24 hours a day, 7 days a week. The eLearning Support Center includes a toll-free telephone number for immediate assistance (1-866-588-3192), email request service, and an online chat service.

- Once you're in the homepage of the class, select from the menu to the left, "Pearson MyLab "

- Click on "Mastering Assignments". You will then be promoted to accept terms and conditions. If you don't have an account with Mastering Physics, select "Create a Pearson Account" and complete the required fields.

- Once you have an account, choose one of the following options:

» Enter a prepaid access code that came with your e-textbook purchase or from the bookstore.

» Buy instant access using a credit card or PayPal.

» Select Get temporary access without payment.

- To access your Mastering assignments later, simply login into Elearning and select "Pearson MyLab" then "Mastering Assignments".

Course Goals & Learning Outcomes: The main objectives of this course are:

- To give students a rigorous introduction into Mechanics in addition to a brief description of wave theory.
- To build personal skills including thinking about science according to the scientific method, critical thinking and problem solving.

Student Learning Outcomes:

1. Add and subtract vector quantities, perform scalar and vector products, determine vector magnitudes and angles relative to a reference frame.
2. Demonstrate how position, velocity, acceleration and time are related mathematically, particularly under conditions of constant acceleration.
3. For 2D and 3D systems, apply position, velocity and acceleration as vector quantities, including situations of circular motion and relative velocity.
4. Understand Newton's three laws relating forces and motion
5. Apply Newton's laws to predict motion for various geometries and for problems involving friction.
6. Understand and use conservation of energy, work, kinetic energy, and power
7. Convert potential energy to force and apply with energy conservation.
8. Interrelate momentum and impulse; understand conservation of momentum; apply momentum to collisions.
9. Understand rotational motion, angular momentum, moments of inertia and how they relate to kinetic energy
10. Understand simple harmonic motion

11.Understand properties of waves such as wave functions, dynamics, power and superposition

What do I need to do to succeed in this course?

This class will focus on developing *critical thinking* and *problem solving skills.*
You will be encouraged to manage your *own success in this course* by participating in all activities and ***by using the tools provided for your success.*** Below is a list these tools

- **Prepare for class:** Lecture summaries will be posted on Elearning including a summary of basic concepts to be covered in class. You are expected to prepare for class by using the lecture summary as a guide. This should prepare you to actively participate in class. CAUTION: *The lecture summaries are not a sufficient learning tool on their own as they only provide a summary* of what you need to know. You are encouraged to supplement with other resources such as (introductory textbooks, Study Area on Mastering, Learning videos, ….etc). Class activities will be planned with student pre-class preparation in mind.
- **Low Stakes Quizzes:** constituting 10% of your final grade. Quizzes will include retrieval exercises and solving simple problems to exercise basic concepts introduced in the lecture notes. They are provided to ***give you the change to build your basic knowledge through practice and to earn easy credit. These quizzes will occasionally include extra credit too.*** *Quizzes* will become available on elearning under "Quizzes and Exams" starting Friday mornings and ***will be due on Mondays by midnigh. No quizzes will be due on exam weeks.***
- ***Class time :*** Student participation is strongly recommended for success in this course.

You may participate by doing two main things:
1. ***Taking notes*** and keeping a record of everything we cover in class: Since our curriculum will be self built and introduced in class, every student is expected to ***take notes and to keep an organized record of everything we cover in class. This will be your best tool*** when ***attempting to solve weekly assignments and will prove very useful when preparing for tests.***
2. Participating in ***problem solving sessions:*** Selected problems will be discussed in class. ***You are strongly encouraged to participate in this peer instruction process by discussing the problem with your group members and attempting to develop a strategy. Be ready to discuss your strategy with the class if needed. This type of activity has been shown to be extremely useful in helping students develop problem solving skills and gain long lasting knowledge.***

- **Mastering Assignments:** Weekly Mastering Assignments will be made ***available each Tuesday after class*** and will be ***due on the following Monday by midnight***. All homework assignments must be completed by the deadline to achieve full credit. ***You may still access the assignments after the deadline to complete missing work for partial credit. You may also***

access them for review until the end of the semester.

Mastering Physics has been proven to be an extremely successful learning tool for students in introductory courses as it provides the feedback necessary for a healthy learning experience. Students are given the chance to *retry attempts* until they get it right. *Hints* can be useful when used with caution. The idea is for you to develop *long lasting knowledge* and *problem solving skills by practicing the concepts learned in class through application to real life situations.*

You are encouraged, though, to use extra caution when solving problems on Mastering and pay attention to the following:

1. Students are encouraged to discuss assignment problems with their peers, TAs or instructor. On the other hand, you are expected to answer the questions individually. This is important since these problems are chosen carefully to prepare you for success in exams.
2. The *biggest mistake is to attempt the assignment problems before you have had a chance to go over the new concepts introduced in class and try a few straight forward examples* (weekly quiz. All textbooks provide solved examples. Make sure you try the examples yourself before looking at solutions).
3. Try not to *guess.* You may find that a certain calculation leads to a correct answer without a good understanding of why. In case this happens, do not move on without understanding the real physics behind the answer. There are many ways you can figure that out.
4. Rushing into using *hints* is specially dangerous if you are working on the problems before you have studied well the new concepts introduced in class.

Students who have followed these guidelines, have done better on exams. On the other hand, not using the Mastering tool properly has *led many students to perform below the level they expected* on exams.

Rubric for the class:

Weekly Quizzes 10 %
Weekly Mastering Assignments 30 %
Three midterm exams with the lowest dropped 20 % each
Final Exam, non-cumulative, Mandatory 20 %

Extra credit
Extra credit on weekly quizzes.
1% for successfully submitting the pre-test ***and*** the post-test
1% for achieving a score > 80 % on the post-test (***for more information on pre-tests and post-tests see the announcement posted on elearning on this subject)***

Exams and The testing Center

The testing center provides a quiet and comfortable environment to help insure our students perform their best with equal opportunities provided for all students.

Our exams will be conducted **at the testing center** and will be available at the testing center the day of the exam. **Every student must reserve a seat at the testing center for the exam no later than 48 hours prior to the exam. You must reserve your seat online through RegisterBlast (exam scheduling tool).**

If you try registering after the 48-hour cutoff deadline, instead of being able to select a time to test you will only see the error message **"All Times are Full"**. You **cannot** schedule more than one exam for one course on the same day. You **can** schedule exams for multiple courses on one day. The testing center does **NOT ALLOW WALK-IN APPOINTMENTS, NO EXCEPTIONS.** So, make sure you ***plan ahead and register for your exam early and in advance.***

***No make-ups will allowed for any reason*. A missed exam is your dropped exam.** More details about exams will be shared prior to exam dates.

Description of Exams

- All of our exams will be Paper exams. A mix of Free response and short answer questions.
- At the testing center, you will be handed a formula sheet together with your paper exam.
- Exams. 1-3 will be 60 minute tests while the final exam will be an open book assignment (more details about the final exam will be discussed in class after the third midterm exam.

Schedule

Schedule is intended to give a brief overview of all material we intend to cover. *The order, date of coverage and time devoted for each section may vary based on need.*

Exam dates are final except in the case of a real emergency situation, which we don't anticipate.

Week starts	Topic
Week1 Jan 20, 22	Introduction, Course Description. Vectors with applications
Jan. 27	End of late registration and last day to add/swap
Week 2 Jan. 27, 29	One Dimensional Motion.
Week 3 Feb 3, 5	Two dimensional and three dimensional motion
Feb 4	Census Day. Last day to drop a class without a "W"
Feb 9-11	Exam 1. Available at testing center Feb 9-11. No class on Feb10

Week 4, Feb 12	Solving Exam 1. Discussion
Week 5 Feb. 17, 19	Newton's Laws of Motion
Week 6 Feb. 24, 26	Applying Newton's Laws.
Week 7 March 3-5	Circular motion.
Week 8 March 10, 12	Work & Energy . Kinetic Energy. Potential Energy & Energy Conservation
March 16 - 22	Spring break. No classes
Week 9 March 23-26	Exam 2. Available at testing center. No class on March 24 & 26
Week10 March 31,April 2	Solving Exam 2. Discussion Introduction to Momentum, Conservation of Momentum & Collisions.
Week 11 April 7, 9	Conservation of Momentum & Collisions. Rotation of Rigid Bodies
Week12 April 14, 16	Rotation of Rigid Bodies
Week13 April 21, 23	Dynamics of Rotational motion
April 27-29	Exam 3. Available at testing center during this period. No class on April 28
Week 14 April 30	Solving Exam 3. Discussion
Week 15 May 5-7	Periodic Motion & Waves. Discussion & Preparation for final exam.
May 10	Final Exam submission. Due at midnight.

All descriptions and timelines are subject to change at the discretion of the Instructor.

Academic Support Resources

https://go.utdallas.edu/academic-support-resources

Student Health Resources: a variety of resources are available to help students to obtain counseling and health care.

Code of conduct and University policies:

- Successful participation in this class is defined as consistently adhering to University and classroom requirements, as presented in this syllabus. Failure to comply with these University requirements is a violation of the Student Code of Conduct
- Academic Integrity: Each student is expected to exercise independent scholarly thought, expression, and aptitude. Copying or assisting in copying of homework assignments or exams, in whole or in part, from the internet, other students or from assignments from other sections/ semesters will be considered to be an act of academic dishonesty, which, once suspected, will be reported to University. Students who violate University rules on academic dishonesty are subject to disciplinary sanctions, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the University, policies on academic dishonesty will be strictly enforced. See more information on under the Student Code of Conduct – UTSP5003. Those students who do not comply will be referred to the Office of Community Standards and Conduct for disciplinary action.
- Disability Services: It is the policy and practice of UTD to make reasonable accommodations for students with properly documented disabilities. However, written notification from the Office of Student AccessAbility (OSA) is required. If you are eligible to receive an accommodation and would like to request it for this course, please contact the Office of Student AccessAbility. This office evaluates the students' needs and provides an assessment. Bring the assessment to your professor. We are committed to meeting every student's needs. Please allow **one week** advance notice.
- In the event of public emergency, inclement weather, etc., that leads to unexpected closure of the university, class will not meet. Please follow the university announcement for its closure and reopening. After the event, look for Announcement on eLearning about the class reschedule.

Comet Creed

This creed was voted on by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:

"As a Comet, I pledge honesty, integrity, and service in all that I do."

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